

Jacques Benzly Te

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LinkedIn: <https://www.linkedin.com/in/jacques-te> | Website: <https://jacquesbte.github.io/myportfolio>

EDUCATION

University of San Diego Shiley-Marcos School of Engineering

BS/BA Electrical Engineering

Minor in Math

San Diego, CA

Expected Graduation, Dec 2026

Major GPA: 4.0 Overall GPA: 3.98

SKILLS

Languages: Python, C/C++, Verilog, System Verilog, MATLAB & Simulink, RoboDK, KUKA Robot Language.

Hardware & Digital Design: FPGA (AMD Vivado, Basys 3), STM32, ESP32, UART/SPI/I2C, PCB Design (NI Ultiboard, Multisim), NVIDIA Jetson Orin Nano (Edge AI Inference).

Analog: Circuit Design & Analysis, Signal Conditioning, Power Systems, PCB Design (NI Ultiboard, Multisim, KiCad), Keysight ADS, Mixed-Signal design.

EDA & Tools: AutoCAD, Librelane, cocotb, Altium Designer, SolidWorks, Lab Instrumentation, Soldering.

Machine Learning & Edge AI: TensorFlow Lite, Edge Impulse, TinyML, IoT (Make.com, API/ Webhooks).

ENGINEERING PROJECTS AND EXPERIENCE

Electrical Engineering Intern | Pacira Biosciences

June 2026 – Present

Supporting next-generation product development for the iovera° cryoneurolysis platform across embedded firmware, analog front-end design, and automated hardware test systems.

- Took ownership of end-to-end design and implementation of an **STM32** based automated solenoid valve test bed for reliability cycle testing, integrating pressure sensing, solenoid drive, and data logging into a unified embedded system.
- Designed a **mixed-signal PCBA** in **Altium Designer** for the analog front end, incorporating instrumentation amplifier **signal conditioning**, linear power regulation, and inductive **load protection circuitry**.

Hardware Engineering Intern, 5G/6G System R&D | Innomountain

Feb 2026 – June 2026

Contributed to RTL design of an MMSE-based signal detection module for a 5G NR massive MIMO beamforming system, focusing on fixed-point datapath design and algorithm-to-hardware mapping.

- Designed a parameterized **Neumann Series MMSE** detectors ($N=8/32/64$, $M=1-64$ MACs) with a systolic array variant in synthesizable Verilog, achieving **100% pass rate** across all configurations at **300 MHz**.
- Achieved **8.1× lower detection latency** vs. previous model and with the **dual-port BRAM** systolic array, delivered an additional **1.63× speedup** with no added memory footprint.
- Scaled banked **RAMB36E2** memory to $N=64$ within **21% BRAM** and **19% DSP** of ZCU106 budget and verified **5G NR** throughput and latency requirements via architecture trade-off analysis.

Performant Manufacturing | Automated Robotic Microfluidic Chip Assembly Cell | Capstone Project

- Led an interdisciplinary team in developing a **KUKA robotic system** for automated **microfluidic chip assembly**.
- Designed and fabricated a custom **EoAT driver PCB** with **KiCAD**, interfacing KUKA digital I/O **with relay and MOSFET-based switching circuits** for end effector control.
- Programmed robot motion paths in **RoboDK** for precision gluing, pick-and-place, and UV curing operations.

CORDIC Trigonometric Processor | 130nm ASIC Tapeout (TinyTapeout, SKY130 PDK)

- Designed and taped out an 8-bit iterative **CORDIC** core in synthesizable Verilog, computing sin/cos over full 360° in Q2.6 **fixed-point** via shift-add rotations and an 8-entry arctan LUT.
- Implemented quadrant folding with sign reconstruction and $1/K$ **gain compensation** for unit-circle output, verified across all 256 input angles against a **MATLAB** golden reference using a self-checking **SystemVerilog** testbench.
- Completed full **RTL-to-GDS** signoff via **LibreLane**, closing timing at 50 MHz in a single $160 \times 225 \mu\text{m}$ tile with zero DRC/LVS errors.

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Wake Word Detection on STM32 Nucleo-L476RG | TinyML Project

- Built a **real-time wake word detection system** achieving **>92% accuracy** under varied noise conditions.
- Trained and optimized a **TensorFlow Lite model with MFCC features**, reducing false activations by **35%**.
- Deployed model on embedded hardware with **<80KB memory footprint** and **sub-150ms latency**, ensuring responsiveness within MCU constraints.
- Deployed the model on STM32 with **<80 KB memory** and **<150 ms latency**, demonstrating efficient **TinyML inference** on low-power MCUs.

Torero Space Program CubeSat Project | Communications & Data Subsystem Lead

- Led a 5-member team to design and validate the **1U CubeSat communications subsystem**, achieving >95% reliable two-way telemetry between the satellite and ground station.
- Integrated and programmed **LoRa (RYLR896)** modules with **ESP32/Raspberry Pi firmware**, optimizing **frequency, addressing, and power for long-range**, interference-free operation.

Fiber Optical Receiver Preamplifier Design | Analog Electronics Project

- Designed a **two-stage BJT preamplifier** (common-emitter + common-collector) for **photodiode-based fiber-optic detection**.
- Achieved **-12.5 k Ω transresistance** gain (**25% above target**) with **linear response** and **polarity control**.
- Delivered **<1% distortion** to a 50 Ω load across 30–60 Ω range, extending -3 dB bandwidth to 11.2 MHz and reducing output impedance from **~1 k Ω to ~100 Ω** (**600% power transfer improvement**).

Configurable Hardware Accelerator for Matrix Multiplication | FPGA Design Project

- Designed a **scalable NxN matrix multiplier** with pipelined MAC units, 6-state FSM, and triple-port memory for high-throughput computation.
- Implemented an **8x8 hardware accelerator** on Basys 3 (Artix-7 FPGA) with UART data loading, LED/7-segment display monitoring, and Python-driven real-time verification.
- Achieved **100 MHz operation performing 512 MACs per cycle**, enabling efficient parallelized matrix multiplication with dynamic data input.
- Optimized FPGA resource usage by leveraging **dual-port BRAM and registered MAC outputs**, extending design scalability up to block RAM limits while maintaining timing closure.

FM Demodulation Using Differentiator-Based Detector | Communication Theory & Signal Processing

- Designed and simulated a differentiator-based FM demodulator in NI Multisim using an **op-amp differentiator, diode envelope detector, and active low-pass filter** for analog signal recovery.
- Implemented and tested the circuit in hardware to convert frequency variations into amplitude variations, accurately recovering an **80 Hz message signal** from a **40 kHz carrier**.
- Optimized differentiator bandwidth (**10 kHz–100 kHz**) and low-pass filter parameters (**$f_c \approx 100$ Hz, gain ≈ 3.3**) to achieve high-fidelity demodulation with **minimal distortion**.
- Validated performance across varying **modulation indices ($\beta \approx 50$)**, achieving **near-perfect waveform reconstruction** and **stable message recovery**.

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DOFTBOT Face Tracking Robot | Jetson Nano Orin Computer Vision & Robotics Integration

- Developed a **gesture-controlled timer** using ESP32 + 5MP camera, recognizing 1–5 fingers with an Edge Impulse ML model.
- Implemented OLED display drivers and countdown logic in embedded C, enabling **fully hands-free timing control**.
- Trained and deployed custom datasets, achieving **95%+ recognition accuracy** and stable inference on constrained hardware.

ML-Powered Finger Timer | Embedded Computer Vision Project

- Developed gesture-controlled timer on ESP32 with 5MP camera and **Edge Impulse TinyML model**, achieving **95% accuracy** recognizing 1-5 finger gestures for hands-free operation.
- Implemented OLED display rendering, countdown logic, and interrupt-driven alerts in embedded C, **optimizing inference and memory usage** within **ESP32's 520KB SRAM constraints**.

AFFILIATIONS: IEEE, ASME, SASE, Tau Beta Pi (Honor Society), USD Torero Space Program, FUSO